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Defining 'the Force' of artificial intelligence in education: exploring the future of teaching through informed speculation

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ABSTRACT

In the 'Star Wars' universe, the Force is a powerful energy that can be harnessed for good or evil. Similarly, artificial intelligence (AI) in education is often depicted as a transformative power capable of revolutionizing teaching and learning. This paper explores possible futures in education in relation to generative AI and predictive AI, using informed speculation to offer insights into the future. As our visions for tomorrow's technology are being defined today, it is important to invite teachers to define the technology we should strive for. This paper presents two co-design workshops involving seven secondary-school teachers with diverse experiences and understandings of AI. Analysis of the workshop recordings informed three narrative episodes that center on lesson-planning in the future, the technical divide, and assessment agency. Education fiction is used as a mode for deep reflection on the results, staging scenarios and exploring the implications of AI in education. The results suggest that while teachers fear changes to their profession, they also offer constructive ideas for AI's potential use. This informed speculation helps to elaborate on challenges and possibilities AI poses for teachers, providing important insights into how to harness AI effectively in teaching while avoiding techno-solutionism by broadening future perspectives.

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Artificial intelligence; AI; AIED; techno-solutionism; EdTech; informed speculation; education fiction

Introduction

Difficult to see; always in motion is the future.
– Jedi Master Yoda (The Empire Strikes Back, 1980)

In the science fiction 'Star Wars' universe, the Force is a powerful energy that can be harnessed for good or evil. The Force, invisible but ever-present, guides and influences people and is wielded by the Jedi (or Sith of the dark side). Similarly, the introduction of artificial intelligence (AI) in education is often compared to 'the Force', with similar attributes of inevitable influence. In this paper, the technology of interest is AI in education (AIED), including predictive AI (PreAI) exemplified by Intelligent Tutoring systems (ITS) that support adaptive learning and automated assessment. It also encompasses generative AI (GenAI), such as large language models (LLMs), chatbots, and image generators like DALL-E. Even though GenAI is not strictly limited to the scope of AIED, this paper defines AIED as encompassing both GenAI and PreAI. The transformative power of these

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technologies is frequently portrayed in policy, and societal discourse as capable of revolutionizing education (Linderoth, Hultén, andStenliden 2024), reinforcing a techno-solutionist discourse where technology is framed as the key to all educational challenges. Amid the current wave of AI hype (Cerratto Pargman et al. 2024; Nemorin et al. 2023) it is reasonable to anticipate that these complex technologies will increasingly engage teachers and influence their work. Furthermore, research cautions against uncritical adoption of AIED as the prevailing enthusiasm may affect teachers' roles in ways that are difficult to predict (Sperling 2024) and raise concerns about invisible labor (Selwyn 2024; Sperling 2024). Moreover, Rahm (2023) argues that even though AIED is often presented as a solution to longstanding educational challenges, teachers are frequently excluded from these discussions. This highlights the need to involve educators in conversations about the implications and implementation of AI technologies in educational settings.

This paper uses speculative methods to engage teachers in reflecting on their future work in relation to AIED. Speculative approaches are used as a methodological tool in two ways. Firstly, in co-design workshops to speculate about the future of teaching, enabling teachers to discuss AI in education beyond today's technologies. Secondly, the results of these workshops are conveyed through narrative episodes grounded in informed speculation. Hrastinski (2023) describes informed speculation as the use of empirical data when speculating on the future. Moreover, Hrastinski and Jandric (2023, p. 510) emphasizes that employing informed speculation is a way 'to abandon the chains of academic formality' and to explore alternative futures we wish to obtain (cf. Costello et al. 2024; Houlden and Veletsianos 2022). Even though it's possible to imagine futures without empirical data, in this study, we emphasize the importance of including teachers when speculating on the future of education.

The paper explores ways of engaging teachers in reflecting on possible futures for education in the age of AIED. The aim is to provide insights that can directly inform decisions made today regarding the teaching practice in relation to AIED. The aim is guided by the question: how might AIED (including GenAI and PreAI) affect teaching roles and practices? As described above, the technology of interest in the context of this paper is GenAI, as well as PreAI in the form of Intelligent Tutoring Systems (ITS), learning analytics and grade prediction integrated in school platforms.

A baseline assumption for this speculation is that Swedish schools will continue integrating technology in classrooms, mirroring trends over recent decades such as the introduction of 1:1 computer programs (Hall and Lundin 2024). This premise provides a grounding reference, ensuring the speculation remains in a galaxy not too far away.

AI in education – situating the study

Introduction of new technology into the school system is not a new phenomenon. Automation of teaching has been sought long before AI entered the educational landscape (Watters 2021), with multiple-choice machines emerging nearly a century ago (Holmes and Toumi 2022). In the 1950s, psychologists such as Sidney Pressey and B. F. Skinner worked on creating various teaching machines. Thus, the novelty of AI should be viewed within the historical continuum of other EdTech innovations. In 1986, Larry Cuban published his seminal work *Teachers and Machines*, tracing the evolution of technology in education from the 1920s onward (Cuban 1986). Cuban describes the introduction of technology as a cycle. Building on Cuban's work, a more recent historical overview by Selwyn (2022a) describes technology in education as a cycle of 'hype, hope and disappointment' (59). The initial phase is characterized by promises of revolution by the introduction of new technology. Nonetheless, two common obstacles to introducing technology in schools are 'the nature "of the classroom and school as work settings" and the "situationally constrained choice" that teachers face when working in schools' (Selwyn 2022a). This leads to phases where hope persists that technology will solve educational challenges, only for it to ultimately fail, resulting in disappointment.

While these historical patterns may be repeating, AIED introduces new dilemmas for teachers. For instance, automation of teachers' work with PreAI requires 'behind the scenes' facilitation by teachers, as noted by Sperling et al. (2022). Furthermore, assessment integrity is especially relevant in relation to GenAI (Concannon et al. 2023; Farazouli et al. 2023), also known for its imperfections and for the ethical dilemmas involved in gathering data from students (Lindell and Utterberg Modén 2025; Yan et al. 2024). Moreover, Williamson (2017) highlights the ongoing 'datafication' of the school system, where assessment, and teaching, is influenced and shaped by digital platforms. This datafication is also discussed by Rensfeldt and Player-Koro (2024) who highlight how the platformatisation of teachers work in, regards to assessment, is forcing teachers to adapt to digital productivity.

The AIED research field presents diverse perspectives. For example, researchers like Rosemary Luckin and Mutlu Cukuvora highlight the potential of AIED (Luckin, George, and Cukurova 2022). Kehoe (2023) also tributes GenAI as potentially revolutionizing by generating lesson plans for teachers, automating and off-loading work effectively. Although, numerous critical scholars also caution against the use of AIED, urging action to avoid techno-solutionism (Perrotta and Selwyn 2019; Selwyn 2022b, 2024; Selwyn et al. 2019).

Education fiction as a means for informed speculation

The ongoing debate about the school of the future reveals sociotechnical imaginaries that encompass both pessimism and optimism. A sociotechnical imaginary is defined as societal views on technologies and affect how new technology is adopted and introduced (Jasanoff and Kim 2015). These imaginaries often align with either utopian views, where AI is seen as a solution to all educational problems, or dystopian, where AI risk deskilling teachers. This mirrors Houlden and Veletsianos (2022), who argue that a focus on dystopia can act as a 'lock-in-function'. To achieve more balanced narratives, they suggest the need for more nuanced portrayals.

Accordingly, drawing on seminal work in critical design (Dunne 1999) and design fiction (Sterling 2005), our study adopts speculative approaches to explore alternative social and cultural realities in education (Galloway and Caudwell 2018). These approaches seek to create new perspectives on complex problems and open space for discussing alternative futures (Raby and Dunne 2013; Wood 2024). For this reason, education fiction, stemming from social science fiction (Gerlach and Hamilton 2003), serves as a platform to explore future scenarios by adopting concepts as if they were real, in an attempt to inform current development of AIED that aligns with teachers' needs and wishes.

Fiction in this context should be 'considered less as a literary genre than as a mode for speculation and reflection' (Gerlach and Hamilton 2003, 168). Hrastinski (2023) explains the term 'education fiction' by highlighting four main characteristics, namely: It '(1) connects the past, present and future; (2) is both speculative and informed; (3) uses atmosphere to evoke emotions; and (4) encourages reflection' (1). This paper uses these characteristics as a focal point for the narrative episodes that will be presented.

As previously mentioned, Houlden and Veletsianos (2022) suggests writing education fiction that takes into account constructive narratives on the future. They state that the narratives 'need to become a part of enabling the end of such systems' (Houlden and Veletsianos 2022, 618). Furthermore, they could serve as a way of taking control of the technological advancements, suggesting that the future is not set in stone, but rather something that we shape here and now. It is also important, according to Rahm (2023), to keep in mind that the question might not be what solutions there are, but rather 'what "solutions" we can destroy'? (2). To invite teachers into speculating about the future, we devised co-design workshops to explore their needs and wishes regarding AI in education. According to Sanders and Stappers (2014b), co-design can be used as a way of 'making sense of the future' (6). Similar co-design initiatives have been conducted in other domains to promote imaginative speculation, for example, in sustainable development (Jacobs et al. 2024). The

following section outlines our methodology, detailing the design and execution of the co-design workshops, along with the analytical process and development of narrative episodes.

Method

As previously noted, this work explores speculative future scenarios for teachers in relation to AIED. To stay grounded in the realities of Swedish classrooms and avoid purely speculative narratives, we devised co-design workshops. These workshops formed the foundation of our narratives, ensuring that researchers were not the sole contributors to the envisioned futures. This section begins by outlining our approach, followed by the structure and rationale of each workshop. We conclude with our analytical approaches and validation steps.

Co-design approach and generative techniques

Involving stakeholders at various stages of the design process is a well-established strategy for creating solutions more closely aligned with users' needs and desires. Commonly referred to as co-design, this approach takes various forms, from participatory design to co-creation (Sanders and Stappers 2014a). It has become a standard practice across different fields, including healthcare and architecture (Luff et al. 1993), marketing (Prahalad and Ramaswamy 2004), and education (Kelly et al. 2019). Co-design in education represents a collaborative approach that bridges gaps between theory and practice by involving both educators and researchers in the development of educational technologies (Penuel, Roschelle, and Shechtman 2007). Sanders and Stappers (2008) emphasize that co-design is rooted in a 'designing-with' mindset and aims to empower designers and people not trained in design to work together in the design development process. Co-design enables teachers to contribute their experiences to shaping the tools they use, while offering researchers valuable insights into the role of technology in teaching and learning (Matuk et al. 2017).

Recognizing the importance of involving teachers in the design and development of educational technology, Matuk et al. (2017) refer to key challenges. They highlight that teachers' limited time and specialized collaborative design skills of must be considered to create meaningful opportunities for co-design participation. Adopting proper tools and techniques and tailoring them to each specific partnership is an essential step for a successful co-design process. Generative techniques are known as toolkits for expression and are a category of design approaches that serve as a bridge between the present and the future, allowing individuals who are not formally trained in design to actively participate in the co-design process (Sanders and Stappers 2014b). As Sanders and Stappers describe, these tools facilitate a process that is both reflective and forward-looking, encouraging participants to draw upon their past experiences, present realities, and future aspirations.

Workshops

This paper uses data from workshops as a means for informed speculation. To devise these authentic workshops, we followed a co-design approach and utilized generative tools and techniques to facilitate teachers' engagement. We employed an inductive approach, with each workshop building on the outcomes of the preceding one. We audio-recorded conversations in workshops and collected observation notes for analysis. The following sections elaborate on the details of these workshops.

Participants

Seven secondary-school teachers (that taught students aged 13-16) were recruited through convenience sampling (c.f. Robson and McCartan 2016). The research team invited teachers in the region to participate by sharing project details via email. Then, participants who taught science, social

science, maths, Swedish and/or technology from a variety of schools were invited to join the study. The participating teachers provided written consent to take part in the project through three audio recorded workshops, each lasting around 90 minutes, spread out across the academic year. The participants were also informed that no sensitive personal data would be gathered, and that the research strictly adheres to the Swedish Research Council's ethical guidelines for good research practice (Swedish Research Council 2017).

Workshop 1

The first workshop was designed to lay the groundwork, focusing specifically on the teachers' prior experiences and current practices. Six of the seven teachers in the group attended. To build shared understanding of the project's status and establish a common language between teachers and researchers, participants engaged in a reflective activity before the first workshop, followed by a brief presentation about the project context.

Using tangible materials and providing a blank space – a common method for creating engaging activities in co-design processes (Sanders and Stappers 2014a) – we facilitated teachers' reflective discussion about AI in general and in education. We created a physical board (see Figure 1) with four cells, each highlighting a different aspect of AI in educational contexts that included *teaching with AI*, *teaching about AI*, *learning with AI*, and *learning about AI*. Teachers were invited to discuss the topic and leave notes in each quadrant. Two researchers facilitated the discussion by prompting with supporting questions.

Workshop 2

The first workshop revealed participants' varied emotions and different familiarity with AIED. It also exposed challenges in moving beyond GenAI and assessment integrity in the conversations. While varying familiarity with AI technologies might seem limiting at first glance, it reflects the broader spectrum of AI literacy in education (c.f UNESCO 2024). Therefore, this difference was considered a benefit that could contribute to a better understanding of the teachers' needs and desires. The second workshop thus aimed to deepen reflection by prompting participants to explore

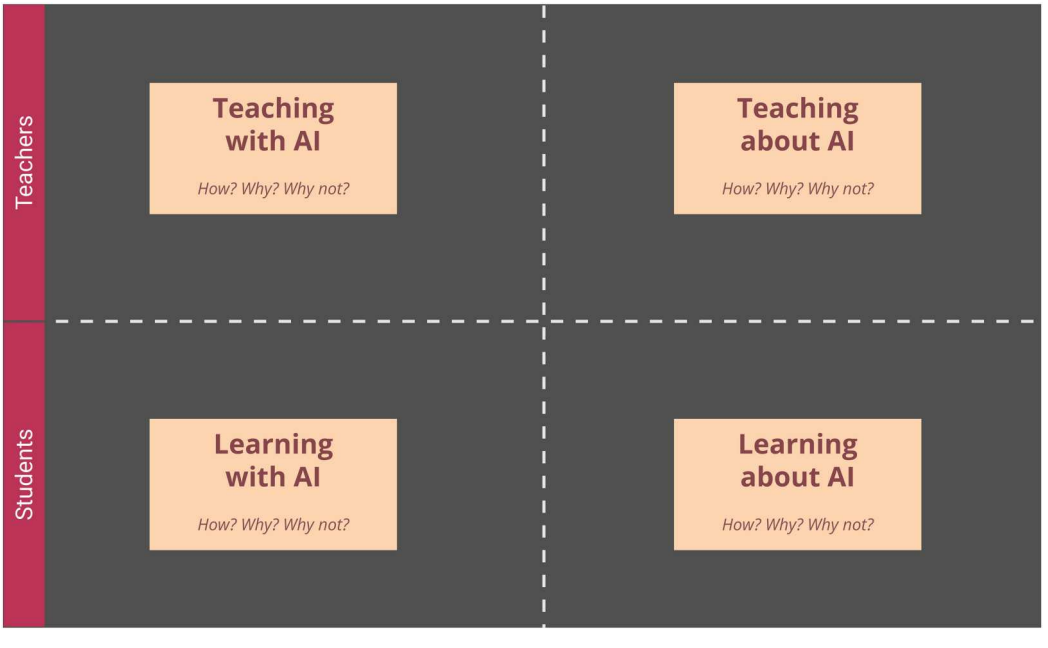


Figure 1. Matrix for workshop 1.

Table 1. Summary of the differences between the original and modified game.

Original game		After modification	
Card type/ number	Description	Card type/ number	Description
Arc x 24	Representing 4 categories of alternative futures in different timespans ranging from a few years to a millennium	Arc x 9	Representing categories of alternative futures in different timespans ranging from 2 to 50 years, selected based on the nature of changes in the EdTech landscape
Terrain x 57	Representing different thematic contexts	Context x 1	Representing teaching and learning with AI as the context
Object x 57	Representing different 'things' from the future	Task x 8	Representing different responsibilities of the teachers, selected based on the outcomes of workshop 1
Mood x 54	Representing the feeling associated with the future scenario	Feeling x 8	Representing the feeling associated with the future scenario, selected based on the outcomes of workshop 1

their needs, fears and hopes for the future. To encourage and facilitate discussions beyond 'here and now', speculation on the future as a tool to understand the present was used as an overarching approach. All seven teachers attended the second workshop.

The main activity was inspired by Candy and Watson's (2015) design fiction game '*The Thing From The Future*', distributed under a Creative Commons (BY-NC-SA) license. By providing prompts, this game encourages players to imagine future scenarios and create an everyday object from that setting. The print-and-play edition of this game includes 192 cards presented in four categories, including arc (representing the future scenario), terrain (representing the thematic context), object, and mood. The game utilizes generative techniques to access players' needs and desires by encouraging them to create fragments of an imaginary future (c.f. Sanders and Stappers 2012). Drawing upon the outcomes of the first workshop, we appropriated the game to our specific context and modified the prompt cards. Table 1 summarizes the final modifications of the original game. After the workshop, the participants had engaged in several different imagined futures. These futures, together with data from Workshop 1, were analysed and used to inform three narrative episodes. The three episodes integrate participants' ideas from both workshops into a cohesive whole.

Data analysis and narrative episodes

The analytical process involved several steps, including validation through research team discussions, seminar feedback from colleagues, and input from participating teachers. The creation of the narrative episodes that form the basis for the results, swerved wse both as an analytical tool by bringing the teachers' ideas to life, and as a pedagogical means of presenting the findings.

Finding 'glow' in the dataset

While coding was essential to the analysis, this study focused specifically on teachers' needs, hopes, and fears regarding AIED, as they could inform the continued development. Therefore, the analysis was guided by identifying instances of 'glow' (MacLure 2013) in the data. MacLure (2013) argues that researchers are not passive observers of data but influenced by it. One such influence is when the researcher 'becomes especially "interested" in a piece of data' (MacLure 2013). In this context, the interview data was first transcribed verbatim by the first author. Passages exhibiting 'glow' were then highlighted using a color scheme in the transcripts, marking quotations that were particularly interesting in regard to teachers' work in relation to AIED. Thereafter, the first author read the transcripts again, and created a mind-map of the passages using a digital mind-mapping tool (miro.com). The mind-map was used to cluster 'glowing' quotations from the workshops. The research team discussed the mind-map clusters suggested by the first author. Accordingly, each

Table 2. Examples of quotations and the clusters that they were grouped in.

Quotation	Cluster in analysis
'We touched a bit on the profession and the future … that there is concern about it because it … assessment, it is ours. It is the professionalism in the teaching practice (…) AI should not solve it for us, because it is actually a craft only we can do'	Teachers' professional expertise / assessment agency
'I would not offload assessment and grading on AI'	
'AI is not with me during the lesson – I have more information for assessment'	
'New technology is like magic because many of us have no ability whatsoever … to understand how it works'	The digital divide
'I don't even know what AI is … It's somewhere there in the can, in the darkness behind the screen, but I don't know what it is and where it lives'	

cluster was further analysed, and its topical name was established. Table 2 illustrates this process by providing examples of the highlighted quotations and the correspondingly formed clusters.

The first excerpts, in the left column of Table 2, highlight the importance of a human-centred approach to assessment in relation to AI – a fear of having assessment offloaded. It expresses questions on teacher assessment agency and was grouped as the cluster 'teachers' professional expertise'. Another passage, that generated interest because it displayed the preconceptions of AI among the teachers is illustrated by the last two excerpts, in the left column. These quotations highlight some sort of resignation – a fear, in relation to new technology and staying abreast of new advancements.

Some interesting aspects were excluded. Examples included participants' discussions about the nature of knowledge and teacher-student relationships in digital classrooms. The clusters that were created at the end focused on the meaning of being a teacher expressed by participants. The final clusters captured teachers' freedom to choose, maintaining control in a fast-paced world, and manifestations of the digital divide in the workplace. The clusters were reviewed by members of the research team for validation before drafting the narrative episodes based on the initial analysis.

Crafting the narrative episodes

In this study, education fiction is not only a creative means to report the results. It is also central to making sense of the imagined futures explored in workshop 2 and the discussions in workshop 1. Furthermore, it offers deeper insight into potential future trajectories by immersing the reader in a fictional school setting. In this way, the data is brought to life through the episodes, serving as both methodological and pedagogical tools.

Writing fiction as research output is a process difficult to make explicit (Cox 2021). However, reflecting on how the episodes were written is essential for transparency. The narrative episodes are based on workshop results. Hence, the feelings they evoke in the narratives reflect those discussed by participants. Each episode represents a cluster from the mind-map analysis of the workshops. Thus, the episodes portray a group of Swedish teachers, and their reflections on artificial intelligence in relation to their profession.

The first author employed a 'what if?' question to effectively guide narrative creation during episode writing (Cox 2021). For example, considering fears of assessment agency: what if the use of AIED significantly reduces teacher assessment autonomy? Contrasting this with 'mundane normality' constitutes a way to construct these narratives (Cox 2021). In the episodes that follow, normality is created by the everyday setting of a Swedish school facing issues and possibilities brought up in the workshops. Since the episodes are built from the analysis of the workshops, they occasionally lean towards either utopia or dystopia, depending on what the teacher group expressed.

The AI software in the episodes is first and foremost described as a school platform, which incorporates elements of generative AI through a large language model which can access both curricula documents as well as teaching material for a seamless lesson-planning experience. It also provides elements of PreAI in the form of ITS personalization by monitoring students' progression and adapts to students' level of learning. As such, the AI school platform is rooted in AIED that we see in schools today, which is a result of the gathered empirical data, where teachers use their current technological understanding to speculate on future trajectories.

Validating episodes with participants

To validate the narrative episodes, participants were asked to read and comment on the initial draft. The first author highlighted sections in the narratives directly drawn from the analysis and posed follow-up validation questions. These included: 'Do you find the narratives believable, in light of your discussions in workshop 2?' and "One aspect that emerged in your discussions was the importance of teachers 'owning' assessment. Based on the narratives, and discussions we had during workshop 2, freely reflect on how assessment in relation to AI could affect your profession and impact you'. Three of seven teachers participated in validation, offering minor suggestions that were incorporated into the narratives.

Results

The narrative episodes in this section are a series of events within the same fictional context. Unlike education fiction communicated by Cox (2021), Hillman, Rensfeldt, and Ivarsson (2019) and Selwyn et al. (2019), who present several different stories and characters, this paper follows the same three main characters. Each episode is first presented with the mind-map from analysis on which the narrative builds. The fictional biology teacher, Amanda, embodies techno-optimism. She uses AI to, among other purposes, generate lesson-plans, something that participants from the workshops discussed potentially helpful future use of GenAI. However, as the teachers also expressed fear about losing autonomy through AIED, the math teacher Henry embodies those fears and underscores the importance of choice in instructional design. A third teacher, John, represents a middle ground, striving to keep up with advancements as the technical divide widens. The fictional teachers thus span a spectrum from techno-solutionism to resistance, creating a tension between acceptance and opposition (Figure 2).

Episode I – lesson-planning in the future

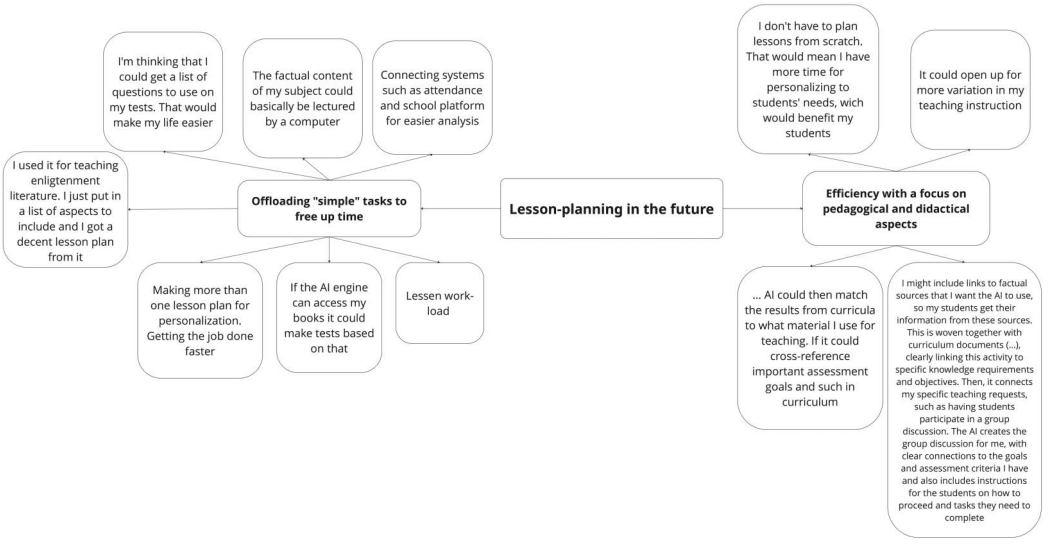


Figure 2. Mind-map for Episode 1 representing the informing data excerpts and created clusters.

Amanda's first day at Nordvik Secondary School was as hectic as she had anticipated. Having just entered the main school building, the corridors were alive with the sound of laughter from teenagers wanting to catch up on their friends' summer endeavours. She clutched her computer to her chest as she nodded to and smiled at students passing her in the corridor. Despite her outward appearance of calmness, her heart raced with the anticipation of the new school that was to be her workplace. She felt the weight of responsibility on her shoulders even before teaching had started, having only yesterday unpacked the last box in her new house. As the sole science teacher in the school

there were no colleagues to lean on for collaboration. However, she was not too worried about it, as she had experience with CloudLINK, which had proved to be very efficient when she had tried it at her previous school. CloudLINK, the AI integrated school platform had been used frequently during her time as a biology teacher in a city much larger than Nordvik. She and her former colleagues had developed a habit of planning (well, *generating* strictly speaking) lessons early in the semester, to have more time to spend on the students and less time planning. The system could cross-reference between textbooks and online teaching material for each subject, seamlessly integrating resources form the Swedish National Agency for Education (such as syllabi and grading criteria). Secondly, it incorporated an adaptive system that predicted the grades for each student by tracking their progress through the academic year. Planning was incredibly easy nowadays, Amanda thought to herself, as she opened the door to the teachers' office.

The door opened as she placed her laptop on the desk with her name on it. A man, who Amanda guessed was a few years her senior, smiled and walked over to shake her hand.

'You must be Amanda!' he said, offering a handshake that was far firmer than she'd anticipated. 'Pleased to meet you'.

'That's me. Thanks' she replied.

The man introduced himself as John, the English teacher. Amanda smiled to herself, thinking that he had a look of someone who would happily stay in on a Friday night to read Virginia Woolfe. Another, younger man made his way over as he entered the room.

'So, you're Amanda? Pleased to meet you. I'm Henry and I teach math mostly, but I've been covering science before you came'.

'We've heard a lot of positive things about you', John said. 'Our principal told us that you are interested in digital resources', he smiled.

'Ah, yes, I mentioned that at my interview', she said and pointed towards her computer. 'I used CloudLINK at my previous school. It works great for lesson plans! I've prepped at least the first couple of weeks already'. Henry raised his eyebrows with a look of curiosity. He looked at John, awaiting his response.

'Wow', John said. 'Honestly, I haven't even used our digital platform that much, but I think I'll have to this year. You know, local policy and all that'. Amanda nodded.

'I heard that during the interview', she responded, shifting her gaze to Henry. 'Do any of the other teachers use it much?' she asked, and Henry shrugged ever so slightly.

'Well, some of our colleagues have', Henry said. 'I've tried it, but like John here I like having books to teach from – it's a matter of preference, I think. I know the end-goal with the AI-plan is to spend less time on paperwork, and more time teaching'. Amanda knew that plan and had learnt that recent years' development in the EdTech industry with AI resources had resulted in fewer teaching positions, simply because teachers could be hired for teaching, and not the other administrative tasks. The local AI-platforms in use made administrative tasks so easy that only one human admin per school district was needed now. Previously, each school would have an admin for support, but fewer humans were involved in education across the country as a whole nowadays (Figure 3).

Episode II – teachers and the digital divide

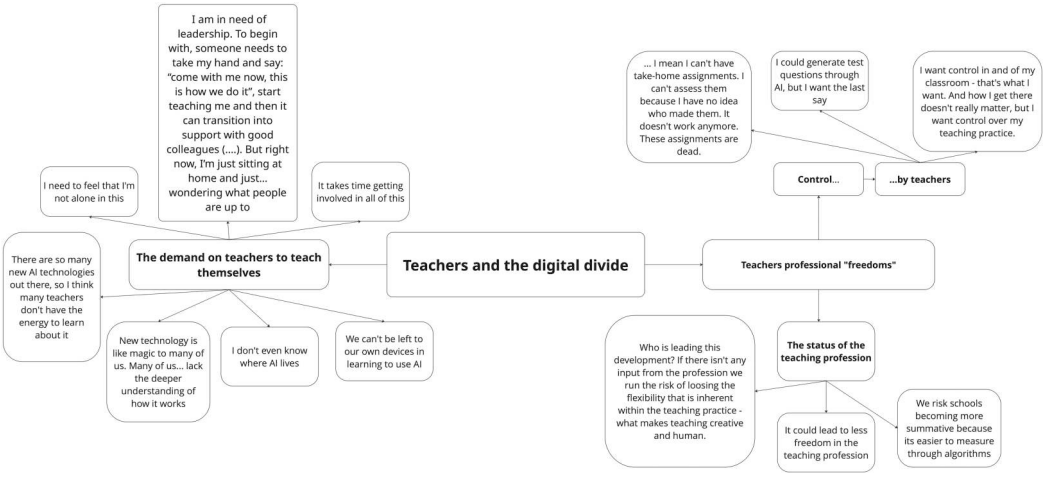


Figure 3. Mind-map for Episode 2 representing the informing data excerpts and created clusters.

Seven months had passed since Amanda started her teaching job at Nordvik Secondary School and she was nothing short of thriving. It was a rainy Thursday in March when Amanda, seated at her desk, watched raindrops race down the window. John, the English teacher, sat behind her, analysing essays by running them through their new AI software – IntelliTech-AI systems. Suddenly, the door opened and Henry, the young maths teacher, rushed in with his computer in hand. He dropped into his chair, setting the computer aside with a heavy sigh, then rested his elbows on his thighs. Amanda glanced at him, troubled.

'Are you ok, Henry?' she offered. John turned around in his chair to look at what was happening.

'Well … ', he hesitated 'honestly, I feel a bit overwhelmed by … everything'. He gestured with his hands toward his desk, where piles of paper lay everywhere. His computer rested on one of the piles. Amanda walked over and saw the program that was open on his computer at the moment: IntelliTech-AI. It had replaced the use of, among other systems, CloudLINK, as it had been favored in the latest procurement by the municipality. Amanda had felt overwhelmed too, by the prospect of not being able to use the tool she had gotten used to but faced no big problems in adapting to the new one. It was, in all honesty, pretty much the same, but with a different interface. Like Cloud-LINK, it offered a large language model for generating teaching material and lesson-plans, while also automating several administrative tasks such as attendance-tracking and scheduling. Furthermore, it also incorporated adaptive learning for the students as well as assessment assistance with predictions on grades based on the students' engagement with the platform. For her it had been intuitive, but for Henry and some of their other colleagues it had been a challenge. There hadn't been any professional development courses, or even a proper introduction this time around. All teachers were expected to be able to adapt to the new AI-system.

'Our principal just called me in to let me know that if I don't start using the software according to our "local guidelines"', he said as he used actual air quotes, 'they will have to take action. Apparently, some of the students have complained that I don't use it'. He let the last sentence out with a sigh.

'What? What do they mean by "taking action"'? John offered as he walked over.

'I'd be mentored by one of the digital lead teachers … mentored! I've been a teacher for five years. I mean, I thought that using AI for administrative tasks would mean more time to focus on teaching quality – but my schedule has just been flooded with more teaching hours and classes. I have no time to actually learn how to use the system. I'm running a race that I can't win'. Amanda didn't know how to reply. She didn't share this reality with him. She did in fact feel that the off-loading the AI software offered had given her the extra time needed to focus on the students who needed her help most. Henry's frustration was very real to her. She just didn't know how to help him (Figure 4).

Episode III – assessment agency and the duel for authority

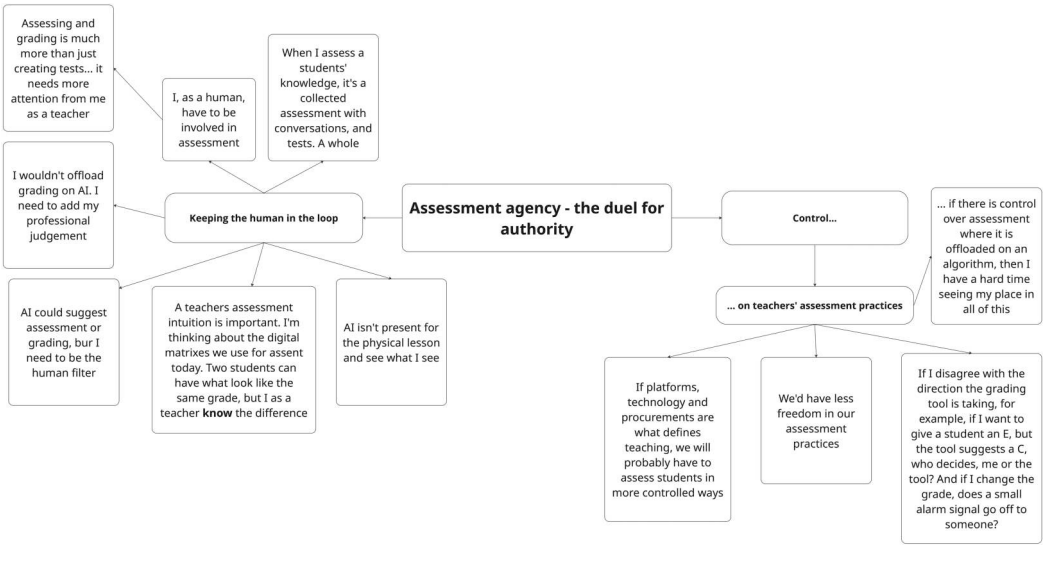


Figure 4. Mind-map for Episode 3 representing the informing data excerpts and created clusters.

The first year of Amanda's teaching job at Nordvik had almost come to an end. Summer was on its way and the birch trees outside slowly bent to the May breeze as Amanda entered the last grades in the system. It wasn't too much of a hassle, as the IntelliTech-AI platform had made preliminary grades in accordance with the assessments made over the course of the academic year. There was, however, a need for some adjustments in the grading after the results of the standardised National Tests. She opened one of her student's learning profiles to write a comment on the results of the National Test and then hovered over the grade preliminarily added by the system. IntelliTech had calculated the accumulative results as a C. However, Amanda knew that the student's latest laboratory tasks, and the National Test indicated a D. She moved the cursor over the grade to correct it, but soon realized she was not able to. A padlock symbol flashed over the screen as she clicked the grade again. A message appeared as she tried a third time: 'Access denied. Deadline for manual grade change passed. Please consult the National Agency for Education'. Amanda frowned. She'd met no such issues in her use of the system before. As she tried yet again, meeting the same results, Henry entered the room followed by their principal. They were in the middle of conversation, but Amanda's frustration got the better of her and she interrupted them.

'I'm sorry, but why am I unable to adjust the grades on the platform?' she pointed to her screen, looking at the principal.

'Oh, well, the deadline for manual changes has passed. We sent out information about this a while ago. The National Agency for Education wants to minimize administrative tasks for teachers, so grading is being automated. Did you not get the email?' Amanda shook her head.

'But … I have to use all data available, and conversations and laboratory tasks in my classroom cannot all be added after every single lesson. I have to be able to adjust the grade at the end of semester!'

Her principal shrugged. 'I'm sorry, but the grading system indicates the grade the student should have according to available data. We have to make sure all data is there in the end'.

'That would require a lot more work than changing the grade now, surely?' Henry jumped to her defence.

'We'll have to look into our routines, but the grade in the system is the grade we finalize' their principal answered.

Discussion

This paper has explored ways of engaging teachers in reflecting on possible futures for education in the age of AIED. The overall aim is to provide insights that can directly inform decisions made today regarding the teaching practice in relation to AIED, addressing the question: how might AIED (including GenAI and PreAI) affect teaching roles and practices? To explore this, we engaged teachers in co-design workshops to facilitate informed speculation. The results, presented as education fiction, emerged from analysing the teachers' opinions to create mind-maps that informed the narrative episodes. Below, we discuss the findings conveyed through these episodes.

The results indicate that while teachers identify areas where AIED could be effectively applied (episode 1), they also express a need for support in professional development (episode 2). Additionally, they express fears about the profession changing if assessment is offloaded to AIED (episode 3). The first episode, inspired by the teachers' ideas on using GenAI to offload 'simple' tasks and enhance efficiency, resonates with current AI discourse where efficiency is seen as a key benefit of GenAI (Concannon et al. 2023; Yan et al. 2024). During the workshops, teachers proposed several ideas, highlighting potential of a GenAI system that could access curriculum documents and teaching materials to streamline their work. These possibilities have also been highlighted by Kehoe (2023), who states that GenAI can revolutionize lesson planning. Notably, the GenAI system imagined by the teachers is intended solely for teacher use, not students. Thus, episode 1 reflects a desire to adopt GenAI that is appropriate to the school context, specifically for teacher support. This envisioned future involves developing school-specific software to offload 'simple' tasks. According to the participating teachers, steps toward this technological future in schools could already be taken.

Episode two elaborates on control within the teaching profession, highlighting teachers' struggle to manage the extend of AIED integration in the new school platform. They express concern over

limited support from school leaders in professional development and guidance for adopting new technologies. In the episode, this is illustrated by an expectation that teachers must take full responsibility for their development without school support. Furthermore, decisions about the available technologies are often made without input from teachers themselves. Instead, local procurements and policy largely dictate the technological infrastructure of the school system, forcing teachers to adapt to available platforms (Rensfeldt and Player-Koro 2024). Player-Koro and colleagues, state that: ' … technology use in public schools is shaped by a combination of local interests and international corporations working with each other to construct nationally appropriate agendas' (Player-Koro, BergvikenRensfeldt, and Selwyn 2018, p. 683). These concerns are visible in the results, even though there is still hope among the teachers that appropriate professional development and teacher involvement in designing AIED systems might help mitigate them. When considering future technological advancements, we must ask who holds agency in deciding which software is implemented in schools, and who is responsible for designing it in the present. Our results clearly indicate a fear of a widening digital divide among teachers and of being compelled to use software over which they have no influence, echoing the concerns raised by Player-Koro, BergvikenRensfeldt, and Selwyn (2018).

The third episode highlights a core aspect of teachers' work, according to the participants. While GenAI for lesson-planning is highlighted as a possibility, automated assessment is met with great caution in the workshops. In the narrative, this is illustrated by the loss of assessment agency in the PreAI automated assessment platform. Speculating on futures involving reduced control over assessment and less freedom to grade students based on professional judgement, the group clearly concludes that automated assessment is not a desirable future. The teachers' hesitations could further be compared to issues raised by, among other Modén et al. (2021) and Yan et al. (2024) and Farazouli et al. (2023) who highlight the challenges in assessment integrity as well as ethical concerns in relation to student data. In a sense, these fears are tied to fears about professional and academic integrity, (Concannon et al. 2023) and raise questions about the future of teaching if assessment is delegated to algorithms. A possible trajectory of teachers' work in the age of AI is thus linked to a fear of losing core aspects of the teaching profession with offloading onto assessment platforms (Rensfeldt and Player-Koro 2024; Williamson 2017). Returning to Rahm's (2023) question on 'what "solutions" we can destroy'? (2), one such solution viewed more as a hindrance than help by the teachers in this study is fully automated assessment.

In envisioning technology soon available in schools, a continuity from today's standards is often assumed. That is, all students and teachers have their own devices. Though, this vision is not completely clear, as the current state of digitalization in Sweden can serve as an example of. In 2023, Sweden retracted the suggested digitalization plan and the Swedish minister of education declared that Sweden would be transitioning from 'screen time to reading time' (Frisk 2023). However, the EdTech industry is blooming, and new software aimed at schools is introduced to 'help' teachers (c.f. Sperling 2024; Sperling et al. 2022). Arguably, this tension between digital and analogue might affect teachers' willingness to adopt AIED in the future. By positioning teachers as active agents in defining digital futures in schools, co-design ensures they remain central to both design and decision-making.

In seeking to return control to the teachers in this study, the validation and feedback stage following the construction of narrative episodes was crucial. Involving teachers in validation ensured their perspectives were accurately interpreted and reflected in the final version of this paper. Participatory approaches like this underscore the value of co-design and co-validation, which, as Rahm (2003) argue, can empower educators by granting agency over challenges faced in today's schools. Speculative and collaborative methods may foster deeper reflection on the sociotechnical pressures within educational systems, ultimately supporting teachers' professional agency (c.f Matuk et al. 2017).

Despite these strengths, the study has limitations which should be noted. Participants taught different subjects, which may influence the generalizability of the findings, as subject-specific

examples were not explored in depth. Additionally, the limited number of workshops provides only a snapshot of teachers' understandings and sentiments toward GenAI and PreAI, warranting the need for further investigation with larger, more diverse cohorts. Although the sample size is limited, the depth of collected data provides meaningful contributions into teachers' attitudes, serving as a valuable foundation for future, larger-scale studies. Thus, while recognizing its boundaries, this study offers credible findings that advance the discourse on AI in education.

To conclude, the introduction of AIED metaphorically represents 'the force' within the education system: a 'power' that shifts the balance and dictates the future of teachers' work. Yet, in the fictional Star Wars universe, the force does not control the jedi. Rather, the jedi harness it to their will. Similarly, teachers must become active agents – Jedi – of AIED's development and implementation. The workshop results show that participating teachers have numerous ideas for when and how to use AI, while also voicing fears about profound changes to their profession. Therefore, teachers (Jedi) should define how AIED is introduced into schools. Imagining these futures provides insight into decisions we must make today.

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Generative AI was used on a limited number of sentences in the narrative episodes for language feedback. No other parts of the paper involved the use of generative AI.

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